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Executive Summary

- **Technology Leader** with **19+ years** advancing from hands-on engineering to executive management, driving the delivery of mission-critical financial platforms processing **billions in daily trading volume** across global markets.
 - **Quantifiable Impact:** Led cross-functional teams on a **£6M FX system transformation** at Barclays, delivering **30% improvement in delivery velocity** and **60% reduction in security vulnerabilities**; managed **multi-million-dollar budgets** at Commerzbank, achieving **15% revenue growth** and enabling market expansion through advanced platform rollouts.
 - Known for building and scaling high-performing teams, implementing Agile and DevSecOps practices, and optimizing delivery cycles resulting in **40% increase in throughput** and **99.97% system availability**.
 - **Regulatory & Risk Mindset:** Delivered **100% compliance** in MiFID II, Dodd-Frank regulated environments; promoted a culture of operational precision and risk management with **zero critical production incidents** over multiple audit cycles.
 - Bridges **emerging AI capabilities** with real-world system constraints, ensuring adoption is deliberate, auditable, and aligned with business and regulatory expectations.
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Core Competencies

- **Engineering Leadership:** Optimized processes and \$10M+ budgets, driving 30% faster delivery.
 - **Platform & Architecture:** Migrated monoliths to microservices, scaled cloud-native systems on AWS and Kubernetes, and automated delivery using DevOps delivering 99.97% uptime, daily releases, and 60% fewer vulnerabilities.
 - **Technical Delivery:** Engineered secure solutions with Java/Spring Boot, Python, SQL, GraphQL APIs, event-driven architecture, and Kafka/Solace; built real-time trading, risk, and data pipelines handling billions daily.
 - **Data & Infrastructure:** Optimized global infrastructure with Oracle, PostgreSQL, MongoDB, KDB+, Kubernetes and Docker, enabling high-performance, compliant financial platforms.
 - **Domain Impact:** Shipped FX and algorithmic trading systems fully aligned with MiFID II/Dodd-Frank, sustaining flawless audit results.
 - **Financial Technology Stack:** Leveraged FIX Protocol and low-latency, real-time market data systems to power high-performance trading and seamless connectivity for billions in daily financial transactions.
 - **Enterprise Security Positioning:** Secure platform design, posture governance, and client-aligned advisory
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Research Outputs / Preprints

Duseja, S. (2026). Beyond Remediation: An Irreversibility Threshold for Preventive AI Governance. SSRN (DOI:). <https://dx.doi.org/10.2139/ssrn.6384060>
Also available on Zenodo.
[10.5281/zenodo.18908588](https://zenodo.org/record/18908588)

Applied Artificial Intelligence (AI)

Built production-grade AI systems demonstrating practical application and reflecting systems thinking, governance focus, and operational readiness.

Portfolio Repository: <https://github.com/sushilduseja>

Professional Experience

Barclays | Vice President, Technical Project Manager – Platform Modernization | Singapore March 2023 – March 2025

- Strategic Impact & Delivery:
 - **Platform Modernization:** Led £6M FX system transformation to microservices and cloud-native architecture, enabling 99.97% uptime and 80% faster deployment cycles for 100+ institutional clients.
 - **Performance & Team Leadership:** Built and mentored distributed engineering teams, improving delivery velocity by 30%; instituted technical development programs sustaining high retention and rapid skill growth.
 - **DevSecOps & Compliance:** Embedded automated security testing, OWASP standards, and end-to-end CI/CD, driving a 60% drop in vulnerabilities and ensuring 100% compliance with MiFID II, Dodd-Frank.
 - **Product Excellence:** Delivered scalable pricing and trade execution workflows, enabling seamless multi-region rollout and supporting significant revenue growth across global FX operations.
- Innovation & Operational Excellence:
 - **Automation Programs:** Instituted workflow optimizations and tool integrations that reduced manual effort by 40%.
 - **Metrics & Reliability:** Deployed monitoring frameworks and operational KPIs that tracked user experience, system performance, and supported flawless audit cycles without critical incidents.

Commerzbank | Vice President, Technical Program Manager – Algo Trading Solutions | Singapore September 2017 – July 2022

- Business Growth & Technology Ownership:
 - **Trading Platform Leadership:** Managed the buildout and enhancement of FX front-office and risk systems handling \$500M+ daily volume; delivered new trading products and algorithmic components, driving 15% revenue growth and supporting Japan market expansion via TFX T-Clear.
 - **Team & Process Excellence:** Directed a core team (10 FTEs, 3 contractors), improving team productivity by 25%; transitioned organization through Agile transformation, reducing cycle times by 20% and boosting throughput by 40%.
 - **Operational Optimization:** Implemented FIX protocol bridges and optimized trading adapters for high-frequency strategies, increasing processing speed by 10% and execution speed by 15%.
 - **Risk & Reliability:** Developed real-time risk, position-keeping, and regulatory reporting frameworks, achieving 10% accuracy gains and 100% flawless audit record.

- **Stakeholder Collaboration:**
 - Coordinated cross-functional projects with legal, compliance, and external vendors to support seamless deploys, sustained uptime, and enterprise-wide business transformation.

IRIS Software Inc. | Associate Project Manager | India | 2017 – 2017

- **Client:** Bank of Montreal - DTCC trade confirmation and settlement platform
- Led requirements, analysis, and implementation of platform enhancements for high-volume trade processing
- Reduced application memory consumption by 30% through GC log analysis and systematic code refactoring

Mahindra Comviva | Technical Manager | India | 2016 – 2017

- **Product:** PreTUPS - prepaid recharge distribution and mobile account management platform
- Directed product development, release management, and feature delivery across distributed teams
- Cut AWS cloud spending by 25% through strategic optimization and best practice implementation
- Improved application stability by 15% by identifying and resolving code inefficiencies

R Systems International | Associate Technical Architect | India | US | 2015 – 2016

- **Client:** DMA Inc. Tax Interchange for Retail - enterprise point-of-sale tax solution
- Rebuilt client relationship and trust through consistent delivery and transparent communication
- Established architectural standards, design patterns, and code review frameworks
- Introduced Test-Driven Development (TDD), achieving 40% reduction in production defects
- Diagnosed and resolved performance bottlenecks using JVisualVM profiling tools

JK Tech | Project Lead | India | UK | 2009 – 2015

- **Client:** Specsavers Optical Group - enterprise retail and supply chain systems
- Managed complete project lifecycles from initiation through post-implementation review
- Improved Agile delivery efficiency by 20% through process refinement and team coaching
- Promoted three times in six years: System Analyst → Technical Lead → Project Lead
- Recognized with Star of the Iteration awards (2012, 2013) and Star of the Quarter (June 2011)

Fiserv | Senior Software Engineer | India | US | 2007 – 2009

- **Client:** Fiserv Trade Order Management - institutional trading platform
- Designed and built automated test framework executing 500+ tests in 7 minutes, saving 40% processing time
- Developed proof-of-concept solutions using emerging technologies (OpenESB, Apache ServiceMix)
- Delivered specifications, program changes, and comprehensive technical documentation

JK Tech | Software Engineer | India | UK | 2005 – 2007

- **Internal Products:** iTimer timesheet application and corporate website modernization
 - Introduced automated unit testing frameworks, reducing manual testing effort by one man-day monthly
 - Mastered Agile and iterative development methodologies in production environments
 - Supported QA, UAT, and production stabilization across multiple concurrent projects
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Certifications & Executive Education

AI Strategy & Safety

AGI Strategy, BlueDot Impact 2026
Future of AI, BlueDot Impact 2026
Generative AI for Executives 2024

Cloud & Emerging Technology

Cloud-Native & Multicloud Apps 2022
Blockchain Essentials 2022

Cybersecurity & Governance

ISO/IEC 27001 Lead Auditor 2025
AI Security & Governance 2025
AI Security Fundamentals 2025
Cybersecurity GRC 2024
Foundations of Cybersecurity 2023

Community & Thought Leadership

AI Safety & Governance Community
Active contributor · 2023-Present

Key Differentiators

- **Technical Depth + Business Acumen:** Rare combination of hands-on engineering experience with strategic product leadership, enabling rapid translation of business needs into technical solutions
- **Global Market Expertise:** Understanding of APAC and European capital markets, with proven ability to navigate complex regulatory landscapes
- **Innovation Driver:** Consistent track record of identifying and implementing emerging technologies that create competitive advantage
- **Future-Proofing Mindset:** Early adopter of AI security and DevSecOps practices; continuously scans for and integrates next-gen technologies that ensure platform relevance and resilience
- **Cultural Intelligence:** Successfully led diverse, cross-cultural teams across multiple time zones and regulatory jurisdictions